PGR111, tasks for week 09

Task 1 – Normalization

Note: The assignment is taken from a previous 3-hour exam (re-sit exam 2019).

The table below shows some data related to exam results and associated relevant information:

coursecode	studentnumber	examdate	coursetitle	studentname	examresult	examregistrations
DB1100	11111	2010-10-10	Databaser	Ola Olsen	B	178
DB1100	11111	2012-11-11	Databaser	Ola Olsen	A	17
DB1100	22222	2010-10-10	Databaser	Jens Jenssen	C	178
DB1100	22222	2012-11-11	Databaser	Jens Jenssen	A	17
DB1100	33333	2010-10-10	Databaser	Ida Olsen	A	178
TK1100	11111	2010-09-09	Digital tekno...	Ola Olsen	D	197
TK1100	33333	2010-09-09	Digital tekno...	Ida Olsen	D	197
TK1100	44444	2010-09-09	Digital tekno...	Ola Olsen	E	197

The **primary key** in the table is: (**coursecode, studentnumber, examdate**).

a) Write down the rule for the second normal form (2NF). Then normalize the table to 2NF: Show which tables you end up with, and which primary and foreign keys the tables contain. Describe any prerequisites for your result.

b) Write down the rule for the third normal form (3NF). Then normalize the table to 3NF: Show which tables you end up with, and which primary and foreign keys the tables contain. Describe any prerequisites for your result.

Task 2 – ER modelling

Use pen and paper or your preferred modelling software.

Check the script `PGR111_09_schema.sql` (you can find it on Canvas, and you can also see the last page to view the SQL scripts) which builds up a small database consisting of 2 tables.

Study the script and draw the ER model that matches the script.

Task 3 – SQL

a) Run the script `PGR111_09_schema.sql` in DBeaver / MySQL Workbench. This database contains information about flights (airplane routes) and reservations (bookings for the flights). Explanation of the table attributes:
   - *flightNumber* identifies a flight route.
   - *departureTime* and *arrivalTime* are, respectively, time of departure and time of arrival for a flight.
   - *fromCode* and *toCode* are 3-letter airport codes for the airport the flight departs from and arrives at.
   - For each airport, it's name (*fromAirport*, *toAirport*) and the city where it is located (*fromCity*, *toCity*) is stored as well.
   - *reservationNumber* identifies a reservation.
   - In addition, the *flightDate* the reservation applies to is stored, as well as its *flightNumber*.
   - For a reservation the *name* of the person who reserved the ticket(s), and how many tickets that got reserved (*ticketCount*), is stored as well.
   - *capacity* is the total number of seats on the flight that day. Different aircrafts are often used on a flight, so the capacity can vary depending on the flightDate.

b) Register the following data in the database:

   Routes are to be established between London (HTR, Heathrow) and Oslo (FBU, Fornebu).

   - One flight departs from Oslo at 0930 and is in London at 1040.
   - Another flight departs from London at 1835 and is in Oslo at 2050.

   The following reservations should be registered for these flights: (seats = ticketCount)
   - Oslo -> London 30.12.2022, capacity 130 passengers:
     - Pedersen, 2 seats
     - Olsen, 1 seat
     - Andreasen, 4 seats
     - Karlsen, 1 seat
   - London -> Oslo 31.12.2022, capacity 95 passengers:
     - Pedersen, 2 seats
     - Jonasen, 1 seat
     - Fransen, 1 seat
     - Andreasen, 4 seats

c) You discover that Fornebu (FBU) was closed down many years ago! You will therefore have to change all references to Fornebu. New values should be *Gardermoen* (*OSL*).

Task 4 – Normalization, then modelling

a) Explain what disadvantages this table structure has.

b) Determine all determinants in the tables. Then normalize the tables to 3NF.

c) Draw an ER model for the new, normalized database structure.

```
---------------------------- PGR111_09_schema.sql ----------------------------

DROP SCHEMA IF EXISTS flightReservation;
CREATE SCHEMA flightReservation;
USE flightReservation;

CREATE TABLE flight
(
	flightNumber int NOT NULL,
	departureTime char(4) NOT NULL,
	fromCode char(3) NOT NULL,
	fromAirport varchar(50) NOT NULL,
	fromCity varchar(50) NOT NULL,
	arrivalTime char(4) NOT NULL,
	toCode char(3) NOT NULL,
	toAirport varchar (50) NOT NULL,
	toCity varchar (50) NOT NULL,
	CONSTRAINT flight_flightNumber_pk PRIMARY KEY (flightNumber)
);

CREATE TABLE reservation
(
	reservationNumber int NOT NULL AUTO_INCREMENT,
	flightNumber int NOT NULL,
	flightDate date NOT NULL,
	ticketCount smallint NOT NULL,
	capacity smallint NOT NULL,
	name varchar(50) NOT NULL,
	CONSTRAINT reservation_reservationNumber_pk PRIMARY KEY (reservationNumber),
	CONSTRAINT reservation_flightNumber_fk FOREIGN KEY (flightNumber) REFERENCES flight(flightNumber)
);
```